

AI4MPAS / IMPACTS

OceanAISpinup handoff

Branch **handoff/prototype-e2e** · 10 September 2026

Map a year--~50 QU240 restart to deep T/S at year ~600, and write a **legal MPAS restart**.

Not a weather model. Not GenCast. Prototype first, then the team takes over.

What this branch is

`github.com/daliwang/AI4MPAS · checkout handoff/prototype-e2e`

In git (review this)

`oceanai/` workflow + GNN + writeback

Tutorials 00–06

X vs ML vs Y snapshot (T and S maps)

AI-ready pack *recipe* (~113 MB tar on Lustre)

Not in git

19 GB Kang raw dump

`data/processed/` working tensors

57 MB *.ml.nc files

Checkpoints / `.venv`

`main` still has old notes and PPTX. Take over from **this branch**.

Pilot workflow

Case v3.GMPAS-NYF_QU240 · `python -m oceanai.run_prototype --stage all`

Stage	What happens
prepare	60 pairs, deep mask k=46...60, cell graph, monthly NYF, scalars
qc	Restart OHC vs history AM (sanity)
baseline	Persistence: predict $Y = X$
truth-writeback	Put Y deep T/S into the X template — proves the writer
train	Residual GraphSAGE on 48 train graphs
infer	Write <code>rst.0055-01-01.ml.nc</code>
snapshot	<code>python -m oceanai.qc.snapshot</code> → X vs ML vs Y

Other GPU cluster: copy the **113 MB** pack, not the 19 GB dump.

Model setup

Item	Prototype choice
Mesh	QU240, 7153 cells, graph from <code>cellsOnCell</code> (41018 edges)
Deep mask	<code>refBottomDepth</code> > 2000 m $\rightarrow$ k = 46...60 (15 levels)
Net	Cell-only residual GraphSAGE, 6 layers, hidden 64
Node input	15 T + 15 S + 7 static + 10 monthly NYF = 47
Target	Residual ΔT , ΔS on 15 deep levels (z-scored; add back to X)
Loss	Huber, weights <code>areaCell</code> $\times$ <code>layerThickness</code>
Split	Train 48 (51–54 $\rightarrow$ 601–604) / holdout 12 (55 $\rightarrow$ 605)
Hardware	Login-node CPU for the published demo

NYF cycles; one namelist. F and θ are encoded but not causal on this set.

Predicted vs copied variables

Restart names. History `activeTracers_*` are not used.

Field	v1 role
temperature k=46...60	Predict ΔT , add to X
salinity k=46...60	Predict ΔS , add to X
T / S k=1...45 (shallow)	Copy X
layerThickness	Copy X
normalVelocity + barotropic	Copy X
Mesh, xtime	Copy X (early date)

No ssh in the restart — do not invent it.

Comparison table — holdout January

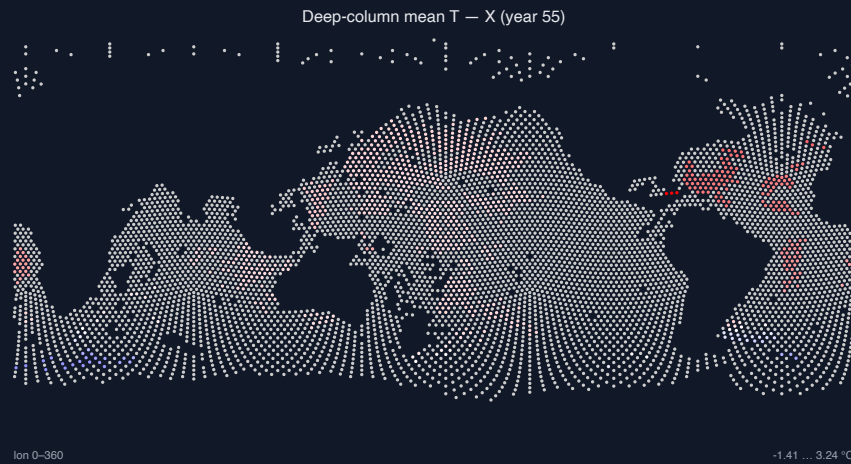
0055-01 (X) vs ML restart vs 0605-01 (Y) · deep, areaxthickness weighted

Metric	X (persist)	ML	Y (truth)
Mean deep T (°C)	1.45	0.10	0.075
Deep T RMSE vs Y (°C)	1.43	0.39	—
Deep T bias this–Y (°C)	+1.37	+0.03	—
Mean deep S	34.736	34.726	34.714
Deep S RMSE vs Y	0.073	0.069	—
OHC 2000 m–bot rel. err. vs Y	0.43%	0.065%	—
Shallow T/S, h, velocity vs X	—	0 (copy)	—

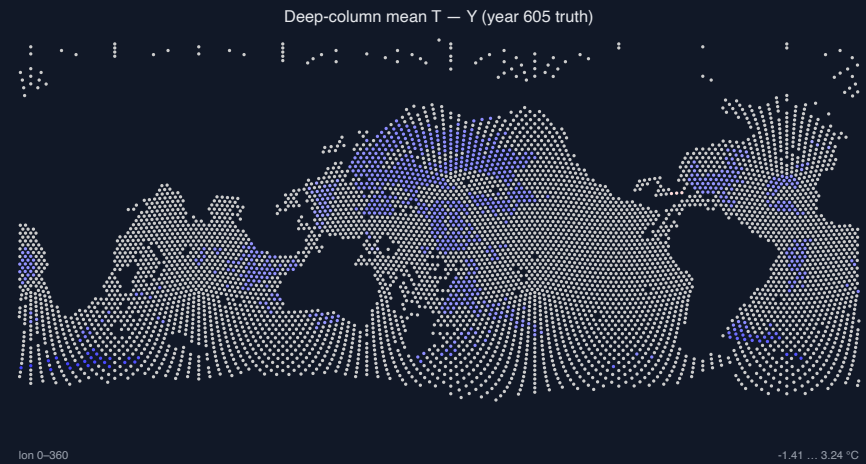
T skill is large-scale cooling. S 550-yr change is small — do not over-claim salinity.

Maps — column-mean deep T

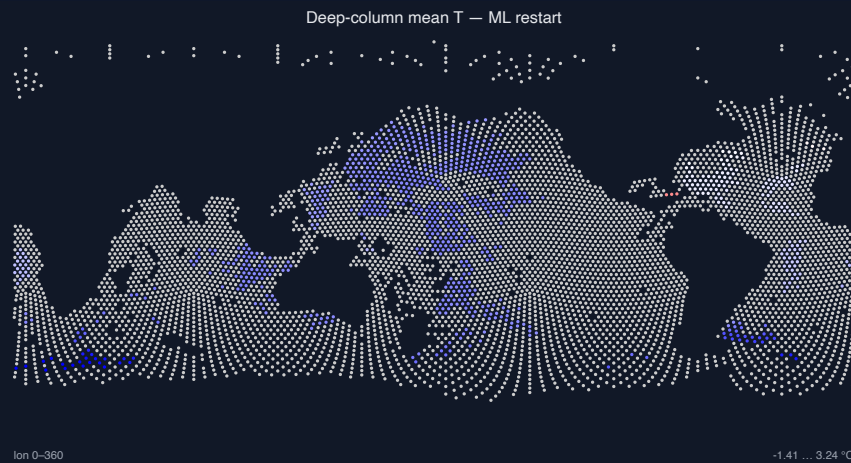
Same color scale. Late ocean is colder; ML follows Y, not X.



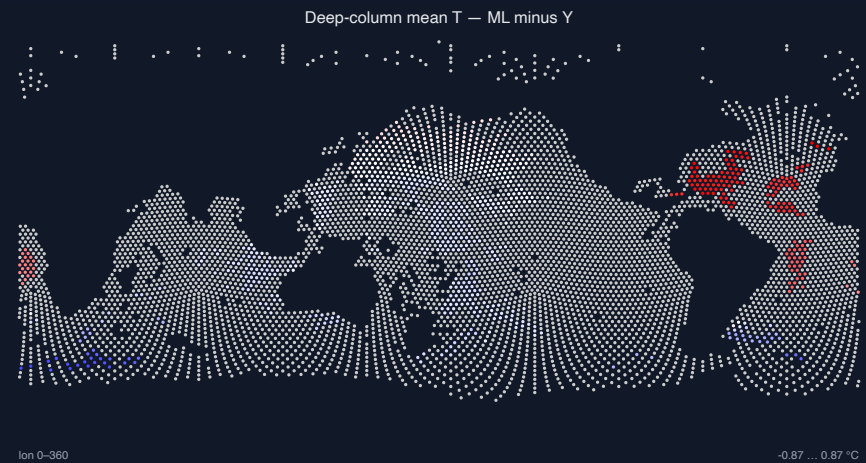
X · year 55



Y · year 605 truth



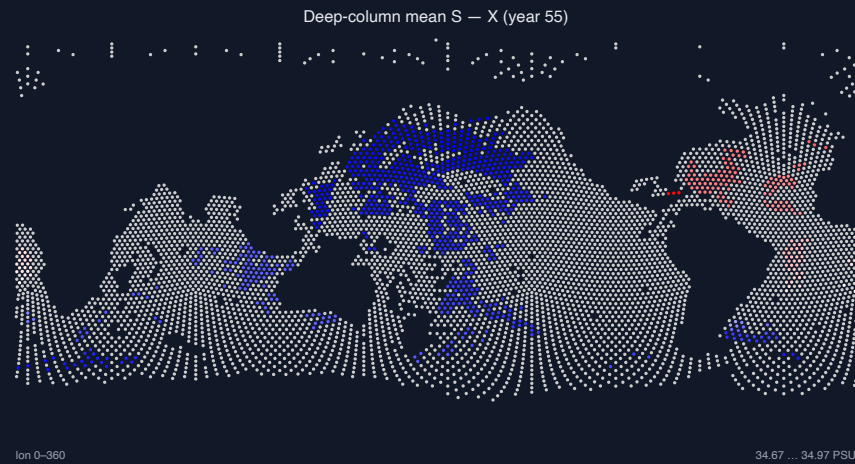
ML restart



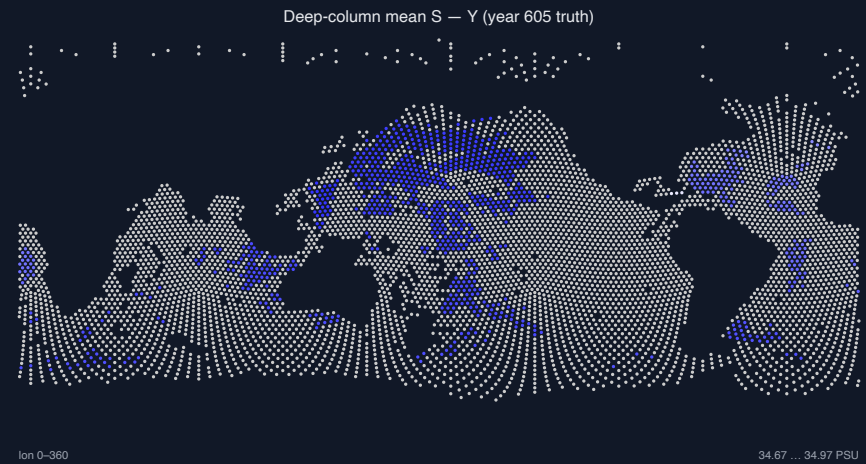
ML — Y

Maps — column-mean deep S

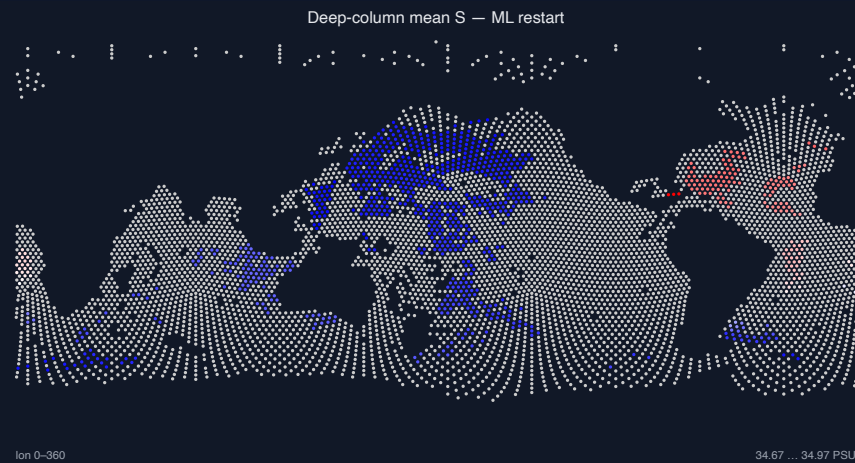
Same layout as T. Global mean X 34.74 $\rightarrow$ Y 34.71; RMSE 0.073 $\rightarrow$ 0.069.



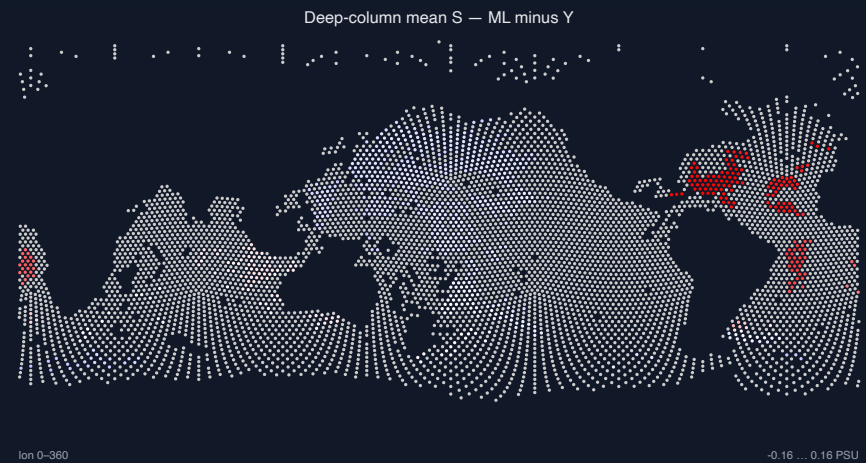
X · year 55



Y · year 605 truth



ML restart



ML – Y

Tutorials

#	Use when you...
00	Need vocabulary (X, Y, deep, persistence)
01	Are on Frontier — smoke run
02	Move tensors to another GPU cluster
03	Touch restart writeback
04	Own OHC / RMSE
05	Will run N-day MPAS (recipe only)
06	Review the X / ML / Y snapshot

Index: `tutorials/README.md`

Who owns what / what not to build

Person	Owns next
Olawale	Pair factory, scalars, training loop
Alice	OHC maps, holdout interpretation
Hyun	N-day forward from *.ml.nc
Dali	Advisor; restart contract

Do not train GenCast on 60 NYF pairs. Do not interpolate T/S to lat-lon. Do not write history tracers into a restart.

Monday morning

```
git checkout handoff/prototype-e2e

# Frontier
python -m oceanai.run_prototype --stage all --smoke
python -m oceanai.qc.snapshot

# Other cluster
export OCEANAI_PROCESSED=/path/to/QU240
python -m oceanai.data.pack_aiready --verify-only
```

Then: `tutorials/00-concepts.md` and `prototype/snapshots/0055-01/compare.md`.